MIL-M-13693A (Ord)
18 July 1962
SUPERSEDING
MIL-M-13693 (Ord)
4 October 1954

MILITARY SPECIFICATION

MINE, APERS, NM, M14E1, WITH INTEGRAL FUZE LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for one type of mine designated as Mine, Apers, NM, M14E1, with Integral Fuze.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification For.
MIL-I-45208 - Inspection Requirements, General Specification for.
MIL-D-45483 - Detonator, Percussion, M46, Loading, Assembling and Packing.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

FSC 1345

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-350 - Jolt Test for Use in Production of Fuzes.
MIL-STD-351 - Jumble Test for Use in Production of Fuzes.

DRAWINGS

ORDNANCE CORPS

82-15-55 - Mine, Apers, NM, M14E1, with Integral Fuze Assembly.
8861849 - Box, Packing, Ammunition, for Mine, Apers, NM, M14E1, with Integral Fuze, Detonator Holder Assembly and Wrench, M22.
8861851 - Carton, Packing, Ammunition, for Mine, Apers, NM, M14E1, with Integral Fuze.
8861850 - Box, Set-up, Packing, Ammunition, Inner and Outer, for Detonator Holder Assemblies.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-7155570 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

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3.2 Mines.-The mines shall comply with all requirements specified on Drawing (dwg.) 82-15-55, and with all requirements specified in applicable specifications.

3.3 Jolt and jumble.-Mine assemblies shall have no explosive elements explode and shall evidence none of the following defects when subjected to the jolt test specified in Standard MIL-STD-350 and the jumble test specified in Standard MIL-STD-351:

a. cracking of the visible surfaces.
b. loosening of the cemented joints.
c. breaking or severe cracking of tetryl ring pellet
d. loosening of the shipping plug (plug turns when an unscrewing torque of not less than one inch-pounds is applied).

The tests shall be conducted as specified in 4.4.1.

3.4 External air pressure.-The mines shall withstand an external air pressure of 3.0 pounds per square inch gage (p.s.i.g.) for not less than 15.0 seconds with no drop in air pressure when tested as specified in 4.4.2.

3.5 Functioning.-The mines shall function satisfactorily, as evidenced by destruction of an aluminum cylinder as shown in Figure 1, when a load of not less than 15.0 pounds nor more than 40.0 pounds is applied when tested as specified in 4.4.3.

3.6 Painting and coating.-The paintings and coatings shall be restored by the loading contractor on any components furnished to him painted or coated when such paintings or coatings have been marred or obliterated.

3.7 Workmanship.-All parts shall be free of chips, scale, dirt, grease, rust, salt deposits, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements

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specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course

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of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 Submission.-The contractor shall submit an initial production sample consisting of fifty consecutively produced assemblies which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the contracting officer. All parts and materials shall be obtained from the same source of supply as will be used in regular production. The initial production sample shall be accompanied by certificates of analyses for all materials. An initial production sample shall also be submitted prior to initiation of production and whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Each mine assembly shall be inspected by the Government for all the inspections specified in 4.3.2.4 and 4.4.3 of this specification.

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4.2.3 Rejection.-If any mine assembly fails to comply with the applicable requirements, the initial production sample shall be rejected and the contractor shall take the necessary corrective action and submit a new initial production sample or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities as necessary until such time as the sample successfully passes the inspection specified or until otherwise directed by the procuring activity.

4.3 Inspection provisions.

4.3.1 Lot formation.-A lot shall consist of mines, loaded and assembled by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Each lot shall contain:

a. parts of one interfix lot number from one manufacturer.

b. detonators of one interfix lot number from one manufacturer.

c. tetryl of one interfix lot number from one manufacturer.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Pellet, ring (see dwg. 75-14-652 covering a detail of dwg. 82-15-55).

Categories	Defects	Method of Inspection	Code No. (see 6.3)
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Weight of pellet	Balance	01001
102.	Pellet chipped or cracked	Visual	01002
Minor:	AQL 1.5 percent		
201.	Diameter of hole, minimum (min.)	Gage	01003
202.	Coating inadequate (bare spots)	Visual	01004
203.	Evidence of poor workmanship (see 3.7)	Visual	01005

4.3.2.2 Detonator holder assembly (see dwg. 75-14-652 covering a detail of dwg. 82-15-55).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Concentricity of detonator with thread	Gage	02001
102.	Detonator disc or cup damaged (punctured or with deep cut)	Visual	02002
103.	Detonator inverted	Visual	02003
Minor:	AQL 1.5 percent		
201.	Length to base flange, maximum (max.)	Gage	02004

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202.	Detonator loose (moves under light finger pressure).....	Manual	02005
203.	Evidence of poor workmanship (see 3.7).....	Visual	02006
204.	Base thread damaged.....	Visual	02007

4.3.2.3 Assembly, less shipping plug (see dwg. 82-15-55).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Ring pellet missing.....	Visual	03001
102.	Firing pin missing or damaged..	Visual	03002
Minor:	AQL 0.40 percent		
201.	Ring pellet cushion missing....	Visual	03003

4.3.2.4 Assembly (see dwg. 82-15-55).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Pressure plate not in safe position (arrowhead not pointing to letter "S").....	Visual	04001
2.	Safety clip missing or insecure.....	Visual	04002
3.	Arrowhead on pressure plate or letter "S" on fuze body missing or illegible.....	Visual	04003
Major:	AQL 0.25 percent		
101.	Shipping plug turns when force of one inch-pound is applied in loosening direction.....	Test	04004
102.	Gap between fuze body and mine body (indicating incorrect assembly).....	Visual	04005

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Minor:	AQL 4.0 percent		
201.	Total height, max.....	Gage	04006
202.	Either cord missing or damaged (worn, frayed or deteriorated).....	Visual	04007
203.	Either cord incorrectly assembled or improperly knotted.....	Visual	04008
204.	Cement missing or inadequate at joint of mine body-fuze body.....	Visual	04009
205.	Grease missing from shipping plug gasket.....	Visual	04010
206.	Shipping plug gasket missing...	Visual	04011
207.	Evidence of poor workmanship (see 3.7).....	Visual	04012
208.	Marking misleading or unidentifiable.....	Visual	04013
209.	Assembly damaged (cracked, chipped or deep cut).....	Visual	04014
210.	Coating damaged (ignoring slight scratches) (see 3.6).....	Visual	04015

4.3.2.5 Carton, sealed (see dwg. 8861851)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Assembly damaged so that contents are exposed or liable to become exposed.....	Visual	05001
102.	Sealing strip incomplete, badly wrinkled, or not in contact with surface for at least 1/2 inch around entire carton.....	Visual	05002

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Minor:	AQL 1.5 percent		
201.	Contents move when carton is shaken.....	Manual	05003
202.	Label missing or peeling (where applicable).....	Visual	05004
203.	Marking or lettering misleading or unidentifiable, including incorrect color code.....	Visual	05005

4.3.2.6 Bag, barrier, sealed (see dwg. 8861851)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Waterproofness of bag destroyed by separation of heat seal, or, rip, tear, or puncture.....	Visual	06001
Minor:	None defined.		

4.3.2.7 Box, packing, wooden, unsealed (see dwg. 8861849)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Carton of mines, box of detonator holder assemblies, or wrenches missing.....	Visual	07001
Minor:	None defined.		

4.3.2.8 Box, packing, wooden, sealed (see dwg. 8861849)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.65 percent		
101.	Hardware missing, broken or loose	Visual-Manual	08001
102.	Strapping missing, broken or loose	Visual-Manual	08002
103.	Contents exposed	Visual	08003
104.	Board broken or split	Visual	08004
105.	DOD symbol misleading or unidentifiable	Visual	08005
Minor:	AQL 2.5 percent		
201.	Hardware improperly engaged	Visual-Manual	08006
202.	Handle missing, insecure, or damaged (deterioration, cut, frayed, etc.)	Visual-Manual	08007
203.	Contents loose (when box is tilted)	Manual	08008
204.	Strapping improperly assembled	Visual	08009
205.	Car seal missing, unsealed, or improperly positioned	Visual	08010
206.	Marking misleading or unidentifiable, including incorrect color code	Visual	08011

4.3.3 Testing.

4.3.3.1 Binder-lubricant content of tetryl (see dwg. 75-14-652 covering a detail of dwg. 82-15-55) Major defect.-The binder-lubricant content of the tetryl shall be determined for each lot of tetryl at the time of loading. A sample of five grams shall be selected just prior to loading and the determination male as specified in 4.4.4. Composite samples shall not be used in this determination. If the binder-lubricant content exceeds the requirement specified, the tetryl shall be rejected. If this determination is made after loading of ring pellets has begun, components loaded therewith shall be rejected. Equipment used in this determination shall be in accordance with 4.3.4. 09001

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4.3.3.2 Jolt and jumble (see 3.3) Major defect-initial production.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, twenty-four loaded mine assemblies shall be selected from each lot for these tests. The mines selected shall be subjected to both tests. The tests shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4. If any mine assembly fails to comply with the specified requirements, the lot shall be rejected. 04016

4.3.3.2.1 Jolt and jumble (see 3.3) Major defect-normal production.-After three consecutive lots have met the criteria of 4.3.3.2, twelve loaded mine assemblies shall be selected from each lot for these tests.

4.3.3.3 External air pressure (see 3.4) Major defect-initial production.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, the sampling plan for this test shall be in accordance with Standard MIL-STD-105 using code letter N and an AQL of 0.25 percent. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4. Any loaded mine assembly which fails to comply with the specified requirements shall be classed defective. 04017

4.3.3.3.1 External air pressure (see 3.4) Major defect-normal production.-After three consecutive lots have met the criteria of 4.3.3.3, the sampling plan for this test shall be in accordance with Standard MIL-STD-105 using code letter L and an AQL of 0.25 percent.

4.3.3.4 Functioning (see 3.5) Major defect-initial production.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, the sampling plan for this test shall be in accordance with Standard MIL-STD-105 04018

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using code letter J and an AQL of 1.5 percent. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4. Any loaded mine assembly which fails to comply with the specified requirements shall be classed defective.

4.3.3.4.1 Functioning (see 3.5) Major defect-normal production.-After three consecutive lots have met the criteria of 4.3.3.4, the sampling plan for this test shall be in accordance with Standard MIL-STD-105 using code letter H and an AQL of 1.5 percent.

4.3.3.5 Check test for deterioration of detonators (see Military Specification MIL-D-45483) Major defect.-If the total time between original acceptance of any detonator lot and the assembly of that lot into the holders exceeds two years, or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in Military Specification MIL-D-45483 immediately before the detonator lot is assembled into the holders. This test shall be performed by the contractor on detonators selected by the Government inspector at the facility assembling the detonators into the holders (see 6.4). 02008

4.3.4 Inspection equipment.-Index of inspection equipment lists number 7155570 identifies available approved inspection equipment designs. The contractor may, however, design alternate inspection equipment in order to facilitate his operation. Designs of any acceptance inspection equipment prepared by the contractor to support his approved written inspection plan (see 4.1.1) shall be approved by the Government prior to fabrication or procuring of such equipment (see 6.2). All acceptance inspection equipment shall be certified by the Government as complying with the approved design prior to its use. Supply and maintenance of inspection equipment shall be in accordance with Specification MIL-I-45607.

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4.4 Test methods and procedures.

4.4.1 Jolt and jumble.-The loaded mine assemblies shall be assembled into a specified test fixture and subjected to the jolt test in accordance with Standard MIL-STD-350. After completion of the jolt test, the mines shall not be disassembled but shall then be subjected to the jumble test in accordance with Standard MIL-STD-351. Visual examination of the internal parts and the tetryl pellet ring using radiographic procedures shall be made of the mine assemblies upon completion of the jumble test. Two pictures of each mine shall be taken; one parellel to, and one perpendicular to its longitudinal axis, in order to determine compliance with 3.3.

4.4.2 External air pressure.-The loaded mine assemblies shall be subjected to an external air pressure of not less than three p.s.i.g. for not less than 15.0 seconds. Observation shall be made for any evidence of a drop in air pressure to determine compliance with 3.4.

4.4.3 Functioning.-The safety clip shall be removed from each loaded mine assembly and the fuze assembly set in the armed position. The armed mine shall then be placed on top a specified aluminum cylinder, 2.0 inches in diameter and 1 1/2 inches in height. Personnel shall then retreat to a safe position behind a suitable barricade. By remote means, pressure shall be applied gradually to the pressure plate at a rate of 1/2 to 1 inch per minute. The pressure applied shall be not less than 15.0 nor more than 40.0 pounds. Observation of the aluminum cylinder shall be made for satisfactory functioning to determine compliance with 3.5 (see figure 1).

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4.4.4 Binder-lubricant content of tetryl.-An accurately weighed portion of five grams (gms.) of the sample shall be transferred to a tared, medium porosity, sintered glass filtering crucible. The sample shall be extracted with four 10-ml portions of acetone, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up, and the sides of the crucible washed down with each 10-ml portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven maintained at 100 degrees to 105 degrees C. for 1/2 hour, cooled in a desiccator and weighed. The above extracting shall be repeated, drying and weighing with individual 10-ml portions of acetone until the loss of weight on subsequent extractions totals less than 0.005 gm. The weight of the residue in the crucible shall be calculated to the percentage of total binder-lubricant in the sample as follows:

$$\text{Total percent of binder-lubricant} = \frac{100\ (A)}{B}$$

where:

A = weight (in gms.) of residue in crucible.
B = weight (in gms.) of original sample of tetryl.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-Packaging for Mine, M14E1, with detonator holder assemblies and arming wrench, M22, shall be in accordance with dwgs. 8861850 and 8861851.

5.2 Packing

5.2.1 Level A.-Packing for Mine, M14E1, with detonator holder assemblies and arming wrench, M22, shall be in accordance with dwg. 8861849.

5.3 Marking.-Marking of the box shall be in accordance with dwg. 8861849.

5.4 Data cards.-Data card information shall be as specified in Military Specification MIL-A-2550.

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6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Submission of contractor-developed equipment designs.- Documentation of contractor-developed inspection equipment designs shall be submitted in duplicate and identified with the associated examination or test required by the specification, to Picatinny Arsenal, Dover, New Jersey, ATTN: ORDBB-ND, for approval. This documentation shall include special equipment designs and a complete description of the use and application of all standard devices and set-ups as well as special equipment designs. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.3 Inspection code numbers.-The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.4 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the detonator assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the detonator assemblies or to the contractor assembling the detonator assemblies into the holders, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the detonator assembly lot was under the jurisdiction of either contractor.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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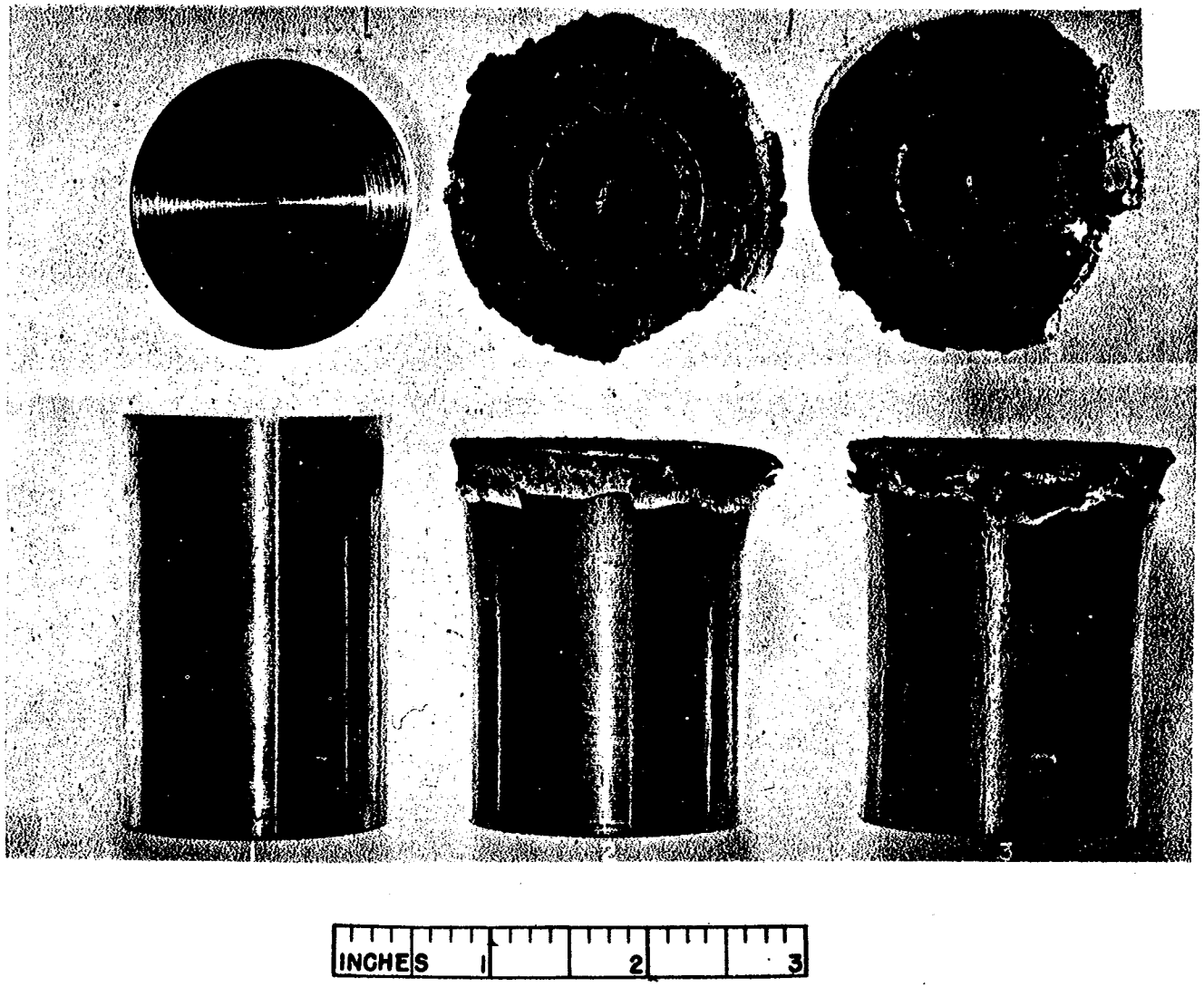


FIGURE 1
Cylinder No. 1: Cylinder prior to test.
Cylinder No. 2 & 3: Showing typical evidence of satisfactory functioning

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